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SUPERSEDING
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May 2, 1960 and
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June 28, 1932

FEDERAL SPECIFICATION

WASTE, MATTED YARNS, (COTTON, COLORED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers clean cotton colored waste yarns intended for use in wiping and as packing in journals.

1.2 Classification.

1.2.1 *Type and grade.* The waste covered by this specification shall be of but one type and grade known commercially as "firsts".

1.2.2 *Bale or package size.* Bales or packages shall be of the following sizes, as specified:

Size 1.—5 pounds.
Size 2.—50 pounds.
Size 3.—100 pounds.
Size 4.—550 pounds.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC-T-191—Textile Test Methods.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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3. REQUIREMENTS

3.1 Material. The waste shall be composed of cotton yarn or a mixture of cotton and rayon yarns conforming to the following physical requirements.

3.1.1 *Length of yarns.* Twenty percent plus or minus five percent of yarns shall be not less than 20 inches in length and not more than five percent of yarns shall be less than 3 inches in length.

3.1.2 *Composition:*

Fine undyed soft spun yarns_40 percent Minimum
Slasher yarns _____ 20 percent ± 2.5 percent
Fine dyed soft spun yarns__40 percent Maximum

Note.—Twenty percent ± 2 percent of nonbright spun-rayon yarns may be used in lieu of the same quantity by weight of cotton yarns.

3.1.3 *Condition.* Yarns shall be free from excessive lint and hard-twisted string, shredded rag, premachined stock, soiled or oily stock, crepe yarn, lustrous rayon, yarns of more than 3-ply, dark-colored yarns, fly sweepings, and dirt or other foreign matter.

3.2 Processing. The waste shall be machined twice to produce a uniformly mixed product with regard to color and length of yarns. The white and colored waste shall be properly floor mixed by spreading each bale of stock material on a laying pile in successive layers so that a portion cut from the edge of the pile to put through the machines will have its proper proportion of every bale of stock material used in the manufacture of the waste. All spools, needles, metal clips, etc., shall be removed when found at any stage of the process.

3.3 Tare and moisture content. The waste shall meet the requirements of table I.

3.3.1 *Tolerance.*

3.3.1.1 *Weight.* A variation of plus or minus 10 percent on the gross weight of the individual bales shall be permitted, provided that the weight of the wrapping, including hoops or ties does not exceed the tare specified in table I. Any excess tare weight between 6 and 10 percent inclusive shall be deducted at the price per pound that is paid for the waste. Tare exceeding 10 percent of the gross weight shall be cause for rejection.

TABLE I.

Gross Weight of [1]Bale or Package	Volume of Bale (approximately)	Tare[1] [2] (maximum)	Moisture Content [1] (maximum)
Pounds	*Cubic Feet*	*Percent*	*Percent*
5	—	6	6.5
50	3.5	6	6.5
100	7.0	6	6.5
550	35.0	6	6.5

[1] See 3.3.1 on tolerance.
[2] Tare exceeding 6.0 percent, but not exceeding 10.0 percent shall be accepted on an adjustment basis (see 4.2.2.1.4).

3.3.1.2 *Moisture.* Where the reported moisture content for a lot exceeds 6.5 percent, but does not exceed 10 percent, payment adjustment for the entire lot shall be made as follows:

Deduction = Total gross weight of shipment multiplied by percentage of excess moisture content multiplied by price per pound.

3.4 Extractable matter. The waste shall contain not more than 2 percent extractable matter when tested in accordance with 4.3.5.

3.5 Absorbency. Water and oil shall be absorbed by the waste within 30 seconds when tested in accordance with 4.3.3.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other in-

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spection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Except when otherwise specified by Civilian Government agencies, inspection shall be in accordance with provisions set forth in Military Standard MIL–STD–105, except where otherwise indicated.

4.2.1 *Inspection of components.*

4.2.1.1 *Testing of components.* In addition to the quality assurance testing provisions of the subsidiary specifications and drawings, testing shall be performed on components and material listed in table II for the characteristics shown therein.

4.2.2 *Inspection of the end item.*

4.2.2.1 *Examination of the end item.* The defects found during examination shall be classified in accordance with 4.2.2.1.1, 4.2.2.-1.2, 4.2.2.1.3 and 4.2.2.1.4. The inspection level (sample size) and acceptable quality level shall be as specified in 4.2.2.1.5. The lot size shall be expressed in packages or bales as applicable. The unit of product for these examinations shall be as follows:

*Unit of Product**

5 pound packages	1.0 lb.
50 pound bales	1.0 lb.
100 pound bales	2.0 lbs.
550 pound bales	10.0 lbs.

Contents of each package or bale to be examined shall be emptied on a clean surface.

* Amount to be taken from each package or bale in the sample size.

4.2.2.1.1 *Overall examination of the waste for defects in cleanness and condition.*

Examine	Defect
Cleanness	Overall uncleanness (appreciably dirty, animal infestation, mildewed, etc.)
	Objectionable odor.
	Contains foreign matter, such as metal particles, wood splinters, paper, dirt, fly sweepings, etc.
	Contains soiled or oily stock.
Condition	Contains yarns other than specified, not free from shredded rag or premachined stock.
	Not free from threads or strings.
	Not uniformly mixed with regard to color and length of yarns and not machined into a homogeneous mass.
	Contains crepe yarn, lustrous rayon, yarns of more than 3 ply or dark colored yarns.

TABLE II.

Component and lot expressed in terms of	Characteristic	Rqmt. para.	Test method	Rqmt. appl. to		No. Det. per sample unit	Results reported as	Smpl. unit	Insp. level
				Sample unit	Lot avg.				
Yarns (100 lbs.)	Length	3.1.1	Std. Comm'l.	—	X	5	To the nearest ¼"	¼ lb.	L–3
	Composition	3.1.2	Std. Comm'l.	—	X	2	To the nearest 0.1%	¼ lb.	L–3

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4.2.2.1.2 *Examination for volume of bale.* Any volume which does not comply with the corresponding volume specified in table I shall be classified as a defect. This examination shall be made prior to the examination indicated in 4.2.2.1.1 (this examination shall not be applicable to 5-pound packages.)

4.2.2.1.3 *Examination for gross weight of bale or package.* Any bale or package that exceeds the specified tolerances of table I or any gross weight less than the weight marked on the bale shall be classified as a defect. This examination shall be made prior to the examination indicated in 4.2.2.1.1.

4.2.2.1.4 *Examination of tare weight.* This examination shall be made subsequent to the examination indicated in 4.2.2.1.1. The total tare weight of the packages or bales taken by sampling shall be obtained. If the total tare weight of the packages or bales exceeds the tolerances specified in 3.3.1.1, the lot shall be rejected. If the total tare weight of these packages or bales exceeds 6 percent but does not exceed 10 percent of the gross weight of the sample, payment adjustment for the entire lot represented by the sample shall be made as follows:

Deduction—Total gross weight of shipment times percentage of excess tare weight times price per pound. Where percentage of excess tare weight = percentage total tare weight minus 6 percent.

4.2.2.1.5 *Inspection levels and acceptable quality level (AQL's).* The inspection levels and AQL's expressed in defects per hundred units shall be as follows:

For defects applicable to	Inspection level	AQL
4.2.2.1.1	I	One class 4.0
4.2.2.1.2	L–3	One class 4.0
4.2.2.1.3	L–3	One class 4.0
4.2.2.1.4	N/A	N/A

4.2.2.2 *Testing of the end item.* Moisture content of the waste shall be performed on each lot in accordance with table III. The inspection level and acceptable quality level shall be as specified in table III. Each sample unit shall be a total of 2 pounds composite drawn from various portions of each package or bale taken for test. One-half of each sample unit shall be accurately weighed to the nearest ⅛ ounce and then immediately placed in a moisture proof container and the cover of the container tightly closed, sealed, properly identified and clearly marked "FOR MOISTURE CONTENT DETERMINATION". The container shall then be submitted to the Government testing laboratory. The remainder of the sample shall be similarly weighed and packaged but retained by the inspector for the umpire test in the event of dispute. If the average results of laboratory tests indicate that the moisture content exceeds 10 percent, the lot shall be provisionally rejected. If the contractor disputes the test results and indicates performance of a retest, the umpire test container shall be submitted to the Government testing laboratory. If the results of the retest indicate that the moisture content exceeds 10 percent, the lot shall be rejected. In all cases, if the reported moisture content exceeds 6.5 percent but does not exceed 10 percent, payment adjustment for the entire lot represented shall be made as indicated in paragraph 3.3.1.2, herein.

4.3 Nonconforming samples. Each package or bale of waste taken for tests shall be tested as described in paragraph 4.3.1 to verify conformance with the requirements of this specification. Any bale of waste in the sample that does not conform to the specification shall be rejected, and if the number of nonconforming bales of waste in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection providing the contractor has repaired or removed all nonconforming bales of waste. The Govern-

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TABLE III.

Characteristic	Rqmt. para.	Test Method	Rqmt. appl. to		No. Det. per sample unit	Results reported as	Inspection level	AQL
			Sample unit	Lot Average				
Moisture content	3.3.1.2	Para. 4.3.2	—	X	2	To the nearest 0.1%	L–2	—
Absorbency*	3.5	4.3.3	X	—	1	Pass or Fail (Describe failure)	L–1	6.5
Extractable matter*	3.4	4.3.4	—	X	2	To the nearest 0.1%	L–1	N/A

* These tests shall be performed prior to the moisture content determinations. The material required for these tests shall be taken from each 2 pound composite sample unit taken for moisture content determination.

ment inspector shall again draw samples when required for both examination and tests from such resubmitted lots to verify compliance with the specification requirements.

4.3.1 *Methods of testing.* The tests shall be conducted in accordance with the following methods of Federal Specification CCC–T–191, and as hereinafter specified:

Identification of cotton—Method No. 1200

Identification of rayon:
Viscose—Method No. 1500
Cuprammonium—Method No. 1520

4.3.2 *Moisture content.* The samples for test shall be accurately weighed and then dried in an oven at 221° to 230°F. (105° to 110°C.) until the weights from two consecutive weighings, taken not less than 5 minutes apart, differ by not more than 0.5 gram of the first of the two weights. The moisture content is the difference between the weight of the waste as sampled (Wo) and the "dry weight" (Wd), expressed as a percentage of the original (sample) weight:

$$\text{Moisture content (percent)} = \frac{Wo - Wd \times 100}{Wo}$$

The average of the moisture content of the individual samples shall be considered the moisture content of the entire lot represented by the samples.

4.3.3 *Absorbency.* The specimen shall be wadded into a ball and laid on a nonabsorbent surface and several drops of water and a light machine oil applied. After 30 seconds the sample shall be examined and the absorption or nonabsorption of the water and oil reported.

4.3.3.1 In cases of dispute, drops 0.04 milliliter in volume shall be applied through a capillary by allowing drops to fall freely into the material from a height of ½ to 2 inches.

4.3.4 *Extractable matter.* A specimen of waste of not less than 5 grams shall be dried to constant weight at 105° to 110°C. (221° to 230°F.) and the weight recorded (B). The specimen shall be extracted with carbon tetrachloride, in a soxhlet apparatus for 4 hours. The carbon tetrachloride shall then be evaporated from the extract and the residue dried to constant weight at 105° to 110° C. (221° to 230°F.). This residue is matter, soluble in carbon tetrachloride and shall be reported as a percentage of the weight of the original oven-dried specimen B.

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5. PREPARATION FOR DELIVERY

(For Civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Federal Standard No. 102).

5.1 Packing. Packing shall be level "C" only.

5.1.1 *Level "C".* The cotton waste shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Shipping containers shall conform to carrier rules and regulations applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, each bale, package or shipping container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Size required (see 1.2.2).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Waste, cotton, not otherwise indexed by name.
Carload minimum weight 20,000 and 30,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Waste, cotton, not otherwise indexed.
Motor volume minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—QM
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